(Please write your Exam Roll No.) Exam Roll No. 00970206117

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] APRIL – MAY 2019

Paper Code: ETAT-202 **Subject: Theory of Machines**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q no.1 which is compulsory. Select one question from each unit. Assume suitable missing data if any.

Q1 (a) What is the function of flywheel? Write at least two application of flywheel. **(2.5)**

(b) Differentiate between Machine and Mechanism. **(2.5)**

(c) Write a short note on degree of freedom? What is redundant link? **(2.5)**

(d) Define the coriolis acceleration with diagram. **(2.5)**

(e) Explain the terms 'fluctuation of energy' and 'fluctuation of speed' as applied to flywheels. **(2.5)**

(f) Draw a tooth profile and mark the following- **(2.5)**
(i) Pitch circle (ii) Addendum (iii) Circular Pitch (iv) Clearance (v) Working depth.

(g) What is radial and offset cam? **(2.5)**

(h) Explain the purpose of dead weight in porter governor. **(2.5)**

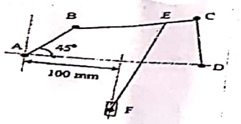
(i) Explain the gyroscopic effect in two wheeler. **(2.5)**

(j) Discuss the causes and effect of vibration in machine? **(2.5)**

UNIT-I

Q2 (a) Explain **any two** from the following:- **(8)**
(i) Whiteworth quick return mechanism
(ii) Scotch yoke mechanism
(iii) Rotary engine mechanism

(b) *In a mechanism shown in the figure below, the link AB rotates uniformly in a clockwise direction at 300 rpm. If the lengths of the links are AB=60mm, BC=160mm, CD=100mm, AD=200mm, EF=200mm and EC=40mm, determine the linear velocity of slider F for the position shown. Also, determine the angular velocity EF.* **(4.5)**



Q3 (a) *In a reciprocating engine, length of crank is 15cm and connecting rod is 60 cm long between centers. When crank angle is 45° from the inner dead center, and crank makes 300rpm clockwise. By using Instantaneous method of center, Find:-* **(7.5)**
(i) Velocity of the piston
(ii) Angular velocity of the connecting rod

(b) Explain klein's construction method for velocity analysis by for a slider crank mechanism. **(5)**

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UNIT-II

Q4 Design a cam profile for operating the exhaust valve of oil engine. It is required to give uniform acceleration and retardation during opening and closing of the valve each of which corresponds to 90° of cam rotation. The valve must remain in fully open position for 30° of cam rotation. The lift of the valve is 50 mm and the least radius of the cam is 30 mm. The follower is provided with a roller of radius 10 mm and its line of stroke passes through an offset distance of 10mm towards the right of axis of the cam. Also find the maximum velocity of follower during ascent period. **(12.5)**

Q5 (a) Draw and explain turning moment diagram for 4 stroke internal combustion engine. **(6)**

(b) State and prove gyroscopic law. **(6.5)**

UNIT-III

Q6 (a) Explain the construction detail and working of porter governor. **(6.5)**

(b) Differentiate between cycloidal and involute profile, with diagram. **(6)**

Q7 (a) State and prove law of gearing? **(6)**

(b) In a machine, power is transferred from one shaft to another parallel shaft through spur gearing the center distance between the two shafts is 600mm and gear ratio is 3. Driver shaft rotates at 120 rpm. Find the number of teeth on each gear, if module is 8mm. What is modified center distance? **(6.5)**

UNIT-IV

Q8 A rotating shaft is attached with four masses A1, A2, A3 and A4 are 200kg, 300kg, 240kg and 260kg respectively. The corresponding radii of rotation are 0.25m, 0.20m, 0.35m and 0.30m respectively and the angle between successive masses are 45°, 75° and 135°. Find the position and magnitude of balance mass required if its radius of rotation is 0.28m. **(12.5)**

Q9 A vibrating system consists of mass of 25 kg, a spring of stiffness 15 KN/m and a damper. The damping provided is only 15% of the critical value. Determine: **(12.5)**
(i) The critical damping co-efficient
(ii) The damping factor
(iii) The natural frequency of damped vibrations
(iv) The logarithmic decrement
(iv) Ratio of two consecutive amplitudes

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